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ECA3102-NANO ELECTRONICS

Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET using NanoHub

A.

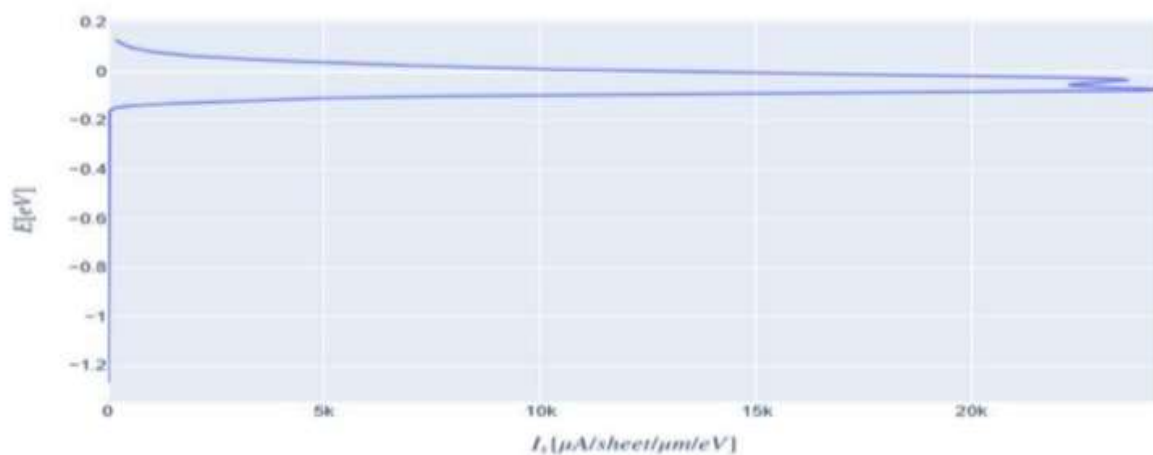
FINFET

Aim:

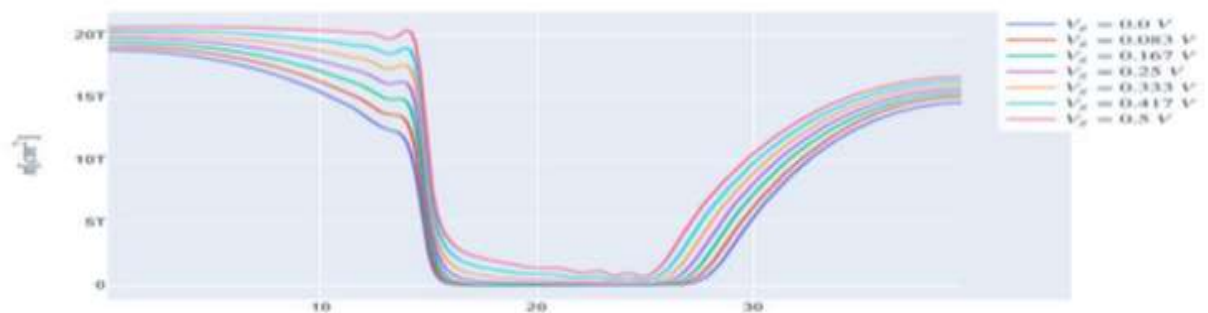
To Obtain the VI Characteristics of FinFET by varying the Gate Channel length and oxide Thickness

Software: NanoHub Online Tube

Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET. Analyze the performance for different Oxide layer thicknesses



Case Study 2: Simulate & obtain the electron density per sheet characteristics of FinFET. Analyze its performance with different Gate region length



OUTCOME:

Thus, the simulation of VI characteristics of FINFET by varying the gate channel length and oxide thickness is analyzed and key learning points are observed.

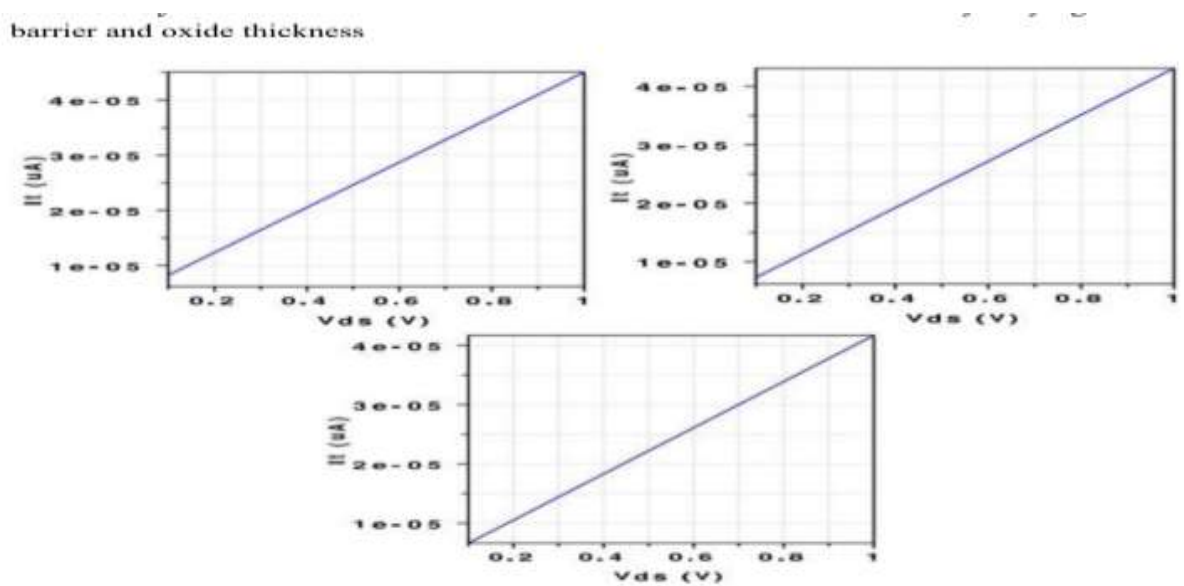
(b) CNTFET

Aim:

To obtain the density of states and conductance of CNTFET by varying the diameter and length.

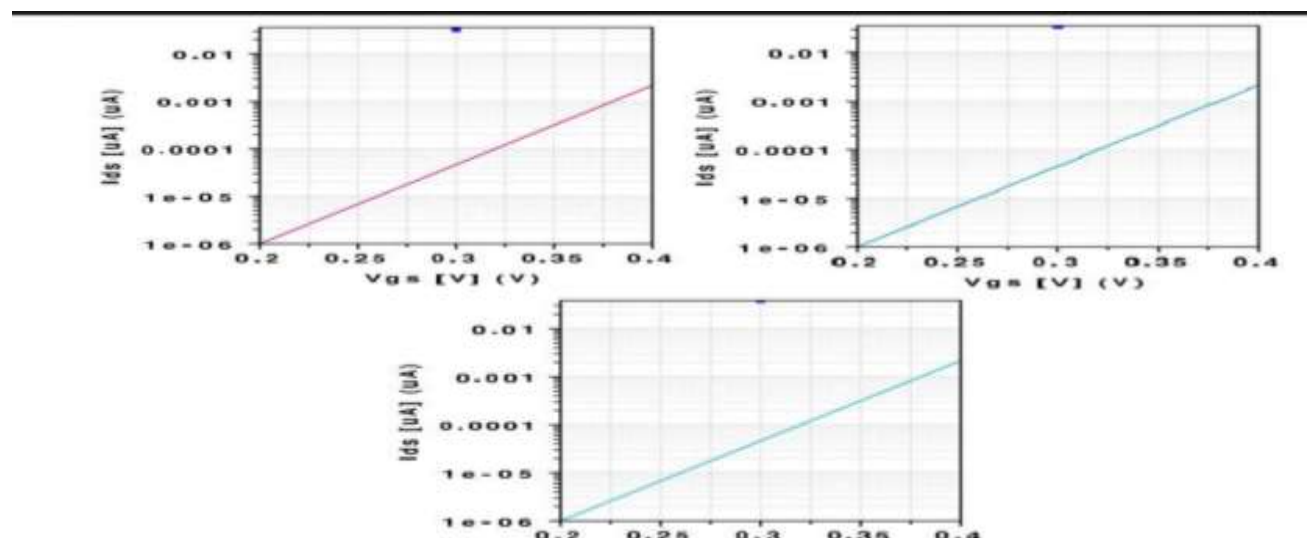
Software: NanoHub Online Tube

Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT

Diameter



Outcome:

Thus the simulation of density of states and conductance of CNTFET by varying the diameter and length is analyzed and key learning points are observed.

(c) GFET**Aim:**

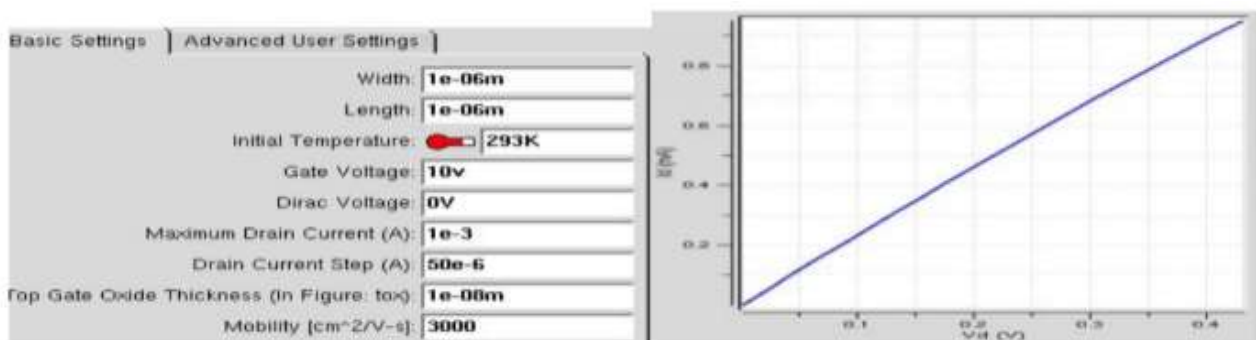
To simulate the Voltage-Current (VI) characteristics of a Graphene Field Effect

Transistor (GFET) using appropriate simulation tools, analyze its performance under

varying gate voltages and drain-source voltages, and understand its behavior as a high- mobility semiconductor device.

Software: NanoHub Online Tube

Case Study 1:How does varying the gate voltage (V_{gs}) affect the drain current (I_{ds}) in a GFET?

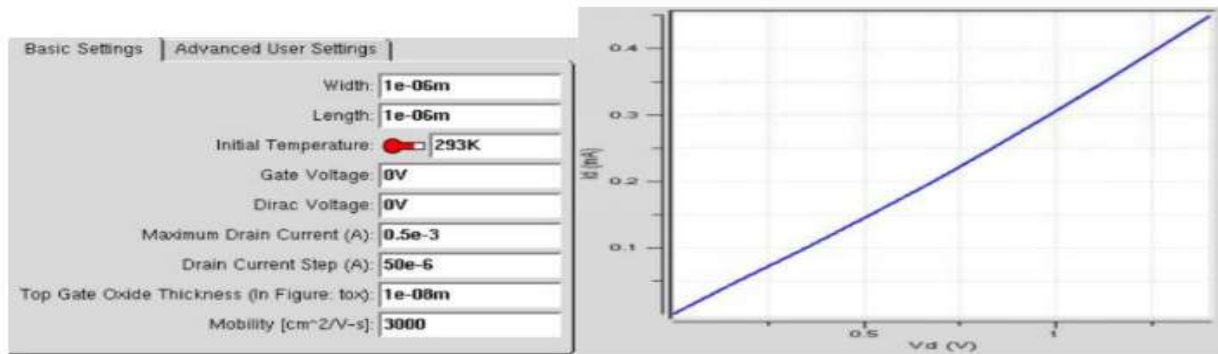
**Inference:For $V_{gs} > 0V$ (Positive Gate Voltage):**

• V_{gs} attracts electrons to the graphene channel, increasing the electron carrier density. • The drain current (I_{ds}) increases linearly in the ohmic region and saturates in the

saturation region as V_{gs} increases further. • Varying V_{gs} directly modulates I_{ds} in a GFET, demonstrating graphene's ambipolar

nature. This behavior distinguishes GFETs from conventional FETs and enables their use in high-frequency electronics and analog circuits.

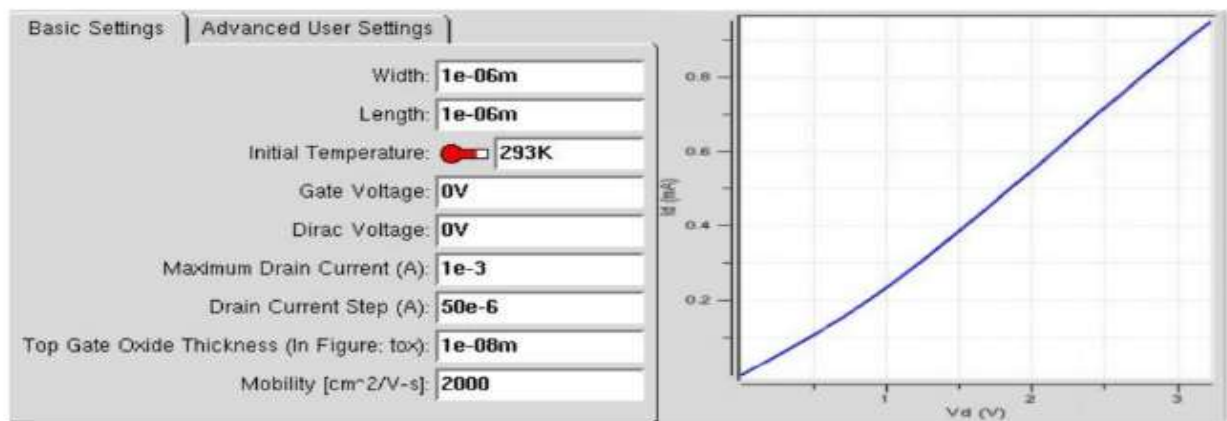
Case Study 2: What factors determine the maximum drain current ($I_{ds,max}$) in a Graphene Field Effect Transistor (GFET)?



Inference:

The VI curve exhibits a symmetric behavior around the Dirac point ($V_{gs}=0V$), where the drain current is at its minimum. For positive V_{gs} , the current is dominated by electrons, while for negative V_{gs} , holes dominate the conduction.

Case Study 3: To investigate how the mobility of charge carriers in a Graphene Field- Effect Transistor (GFET) with a mobility of $2000\text{ cm}^2/V\cdot s$ influences its current-voltage (VI) characteristics.



Inference:

The simulated VI curves show a relatively high drain current (I_{ds}) for a given gate-source voltage (V_{gs}), demonstrating efficient carrier transport in the graphene channel. High carrier mobility ($2000\text{ cm}^2/Vs$) minimizes scattering effects, enabling rapid carrier movement and improved current conduction. Result:

The VI characteristics of the Graphene Field Effect Transistor (GFET) were successfully simulated. The device demonstrated:

- A sharp increase in current with increasing V_{gs} due to high carrier mobility, with the drain current reaching a maximum of 0.5 mA at $V_{gs}=5V$.
- A low threshold voltage of approximately 0.5 V , indicating efficient operation at low

gate voltages. The linear and saturation regions of operation were clearly observed, showing typical GFET behavior.

RESULT

The simulation confirms the potential of GFETs for high-speed and low-power applications in future electronic devices.